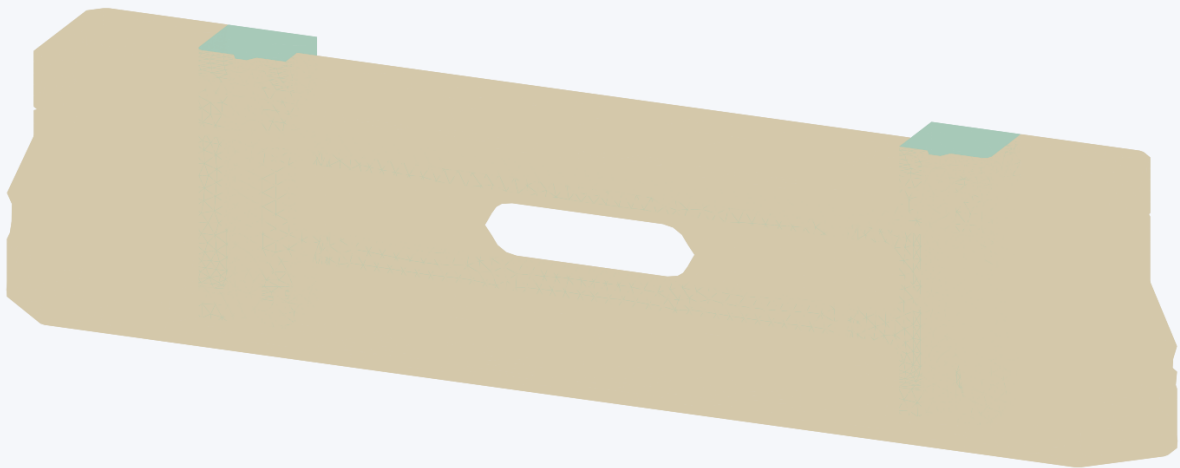


Partial report. Selected-run evidence status: failed

Geometry

Undeformed geometry

Exterior shape from the selected mesh



Mesh-derived exterior • source geometry unchanged

Undeformed geometry from selected mesh. Exterior shape derived from the selected run's mesh; this is not a CAD render.

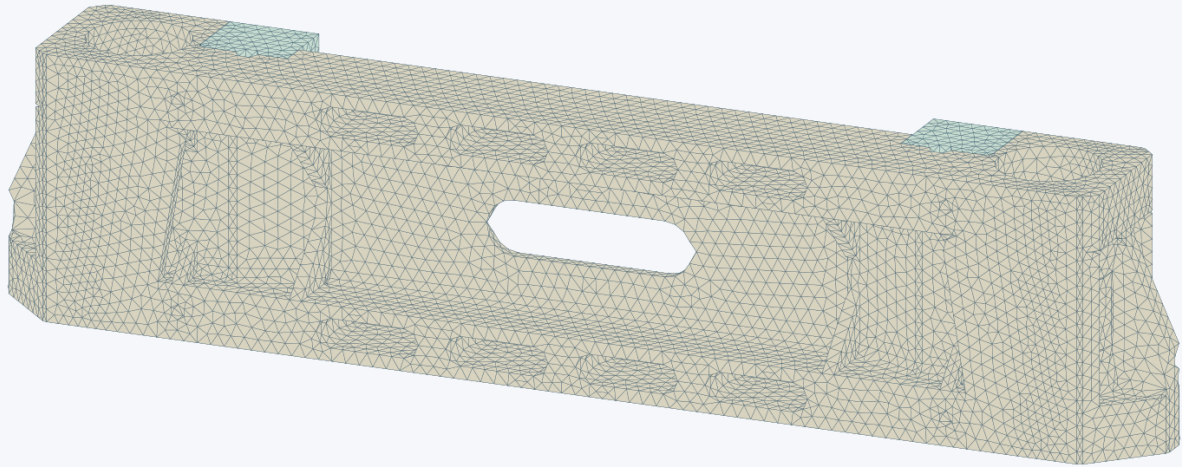
Mesh

Mesh overview and local detail

Mesh nodes	19208	
Tetrahedral elements	67240	
Meshed parts	2	
Nominal element size	4.0	mm
Minimum mean-ratio quality	0.0774	0-1

Mesh overview

Undeformed exterior mesh

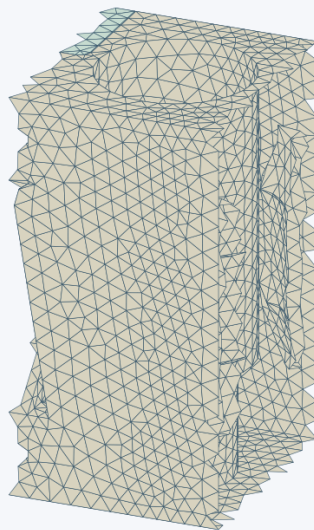


Exterior mesh faces • undeformed geometry

Mesh overview. Undeformed exterior topology of the selected case, verified against its recorded solved-mesh hashes.

Pad-root refinement detail

Pad-root region



Exterior mesh faces • undeformed geometry

Pad-root refinement detail. Positive pad-root region, $X = 112\text{--}150\text{ mm}$. Mesh metadata records a 0.65 target factor (2.6 mm local target).

Mean-ratio quality describes element shape and does not establish stress or time-step convergence.

Setup

Recorded selected-case model

Initial impact velocity	50	m/s
Requested simulation duration	8	ms
Foam depth	152.4	mm
Foam width	41.275	mm
Foam height	152.4	mm
Foam element size	10	mm
Initial contact gap	0.1	mm
Contact friction coefficient	0.1	dimensionless
Foam curve scale	1	dimensionless
Time-step scale	0.6	dimensionless
Requested state-output interval	0.01	ms
Startup mass-scaling DT2MS	-1.5e-07	s
Motor mass	0.1616	kg
Moving mass	0.6503408	kg
Pusher mass	0.4887408	kg
Initial kinetic energy	812.926	J

Recorded material, contact and boundary assumptions

- Body and bolted cable cap flexible; four local perfectly clamped bolt collars. No bolt preload, slip or failure.
- Motor loaded upper-bound inertia distributed over actual bore nodes; no motor housing stiffness or exact inertia tensor.
- PA12-CF10 elastic isotropic proxy $E=1806.6\text{MPa}$ $\nu=.35$ density 1060; print +Z motor axes. No strength/failure claims.
- No fracture, erosion, motor thrust, carrier, released endcap or invented 35g hardware allowance.
- Rigid 8mm aluminum spreaders initially stationary; fixed x,y and rotations, free z without travel limit.
- Front and rear pusher bore-quadrant averages follow fixed zero-clearance coaxial guide axes through eight disjoint LINEAR_GLOBAL X/Y equations; every Z DOF is free. Cable compliance and clearance are

idealized.

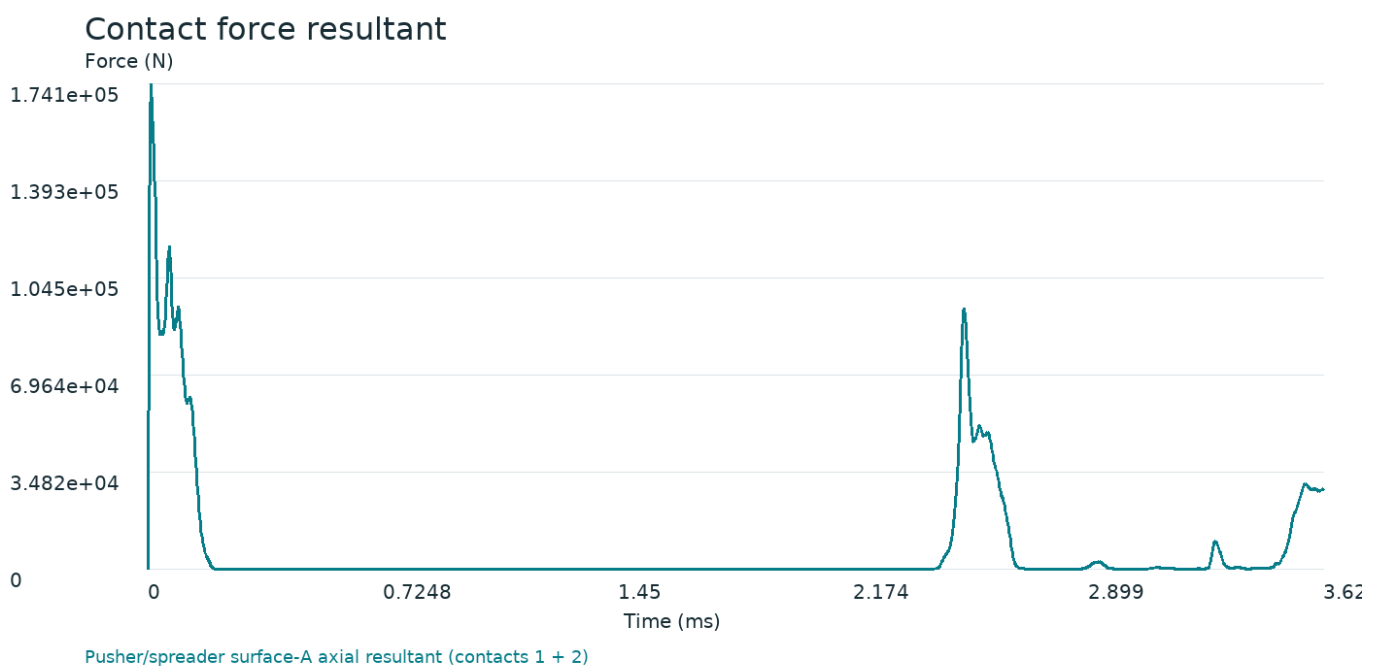
- All axial stopping reaction passes through transient contact; no axial guide constraint.
- Foam rear z fixed, x/y free: frictionless fixed backing; sides unconfined.
- MAT057 synthetic loading and unloading; HU=.05 BETA=0 DAMP=.1 SHAPE=1 FAIL=0. Not a product material model.
- Generic foam anchored at one catalog point; remaining curve chosen for nonbottoming comparative stop.
- Startup-only local mass scaling requested; added mass and inertia error MUST be assessed from solver output.
- Mesh, timestep, energy and momentum require independent assessment.

Solve

Recorded run status

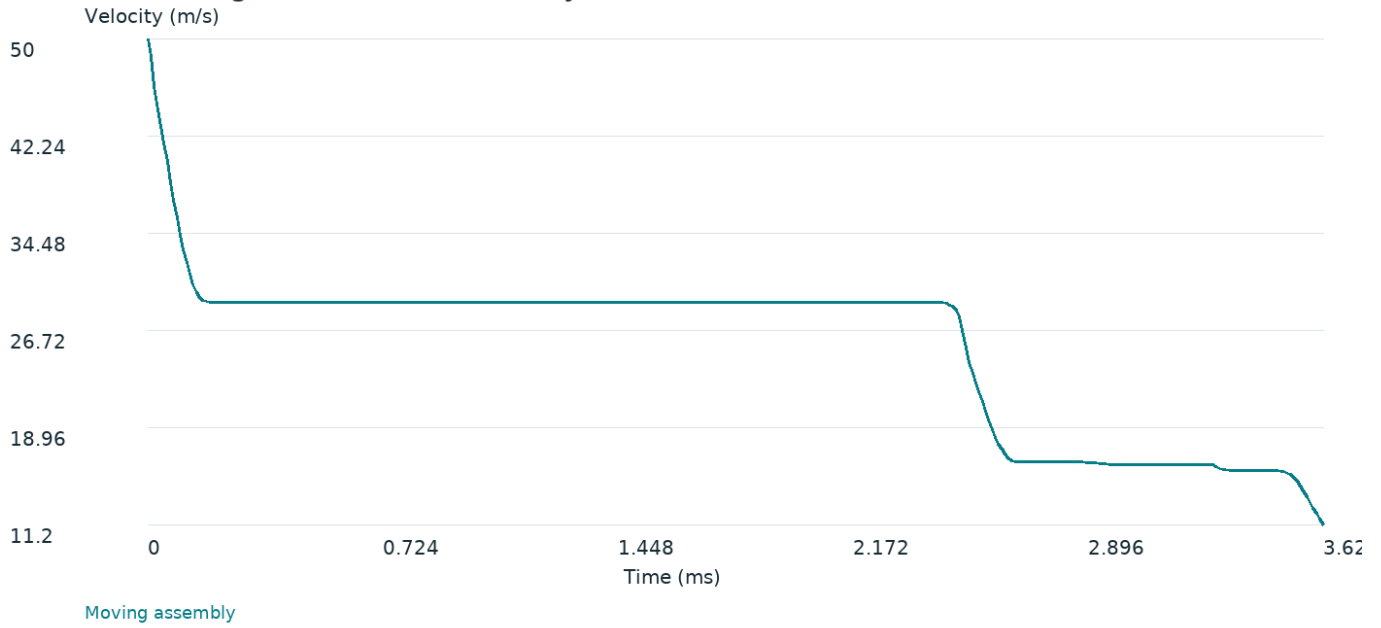
failed

Separate solve charts



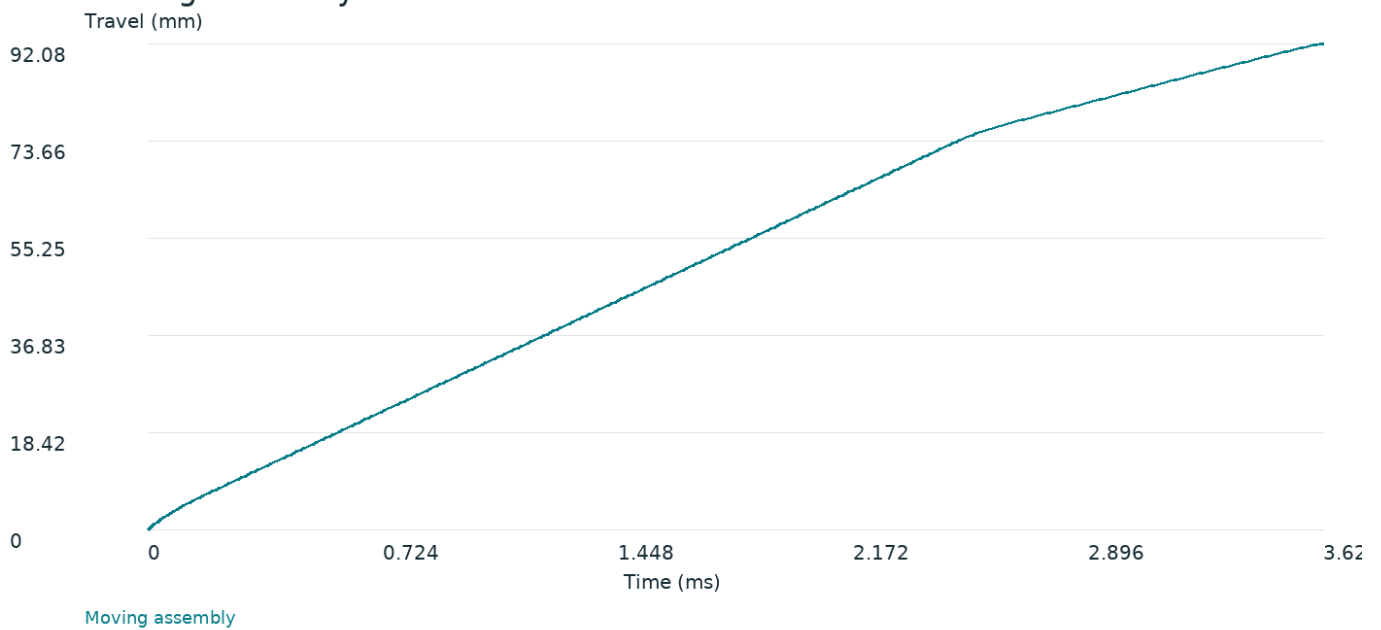
Contact force resultant. projects/compliant_stop_340/cases/depth6-guided-v2/assessment/contact_histories.csv

Mass-weighted forward velocity



Mass-weighted forward velocity. projects/compliant_stop_340/cases/depth6-guided-v2/assessment/time_histories.csv

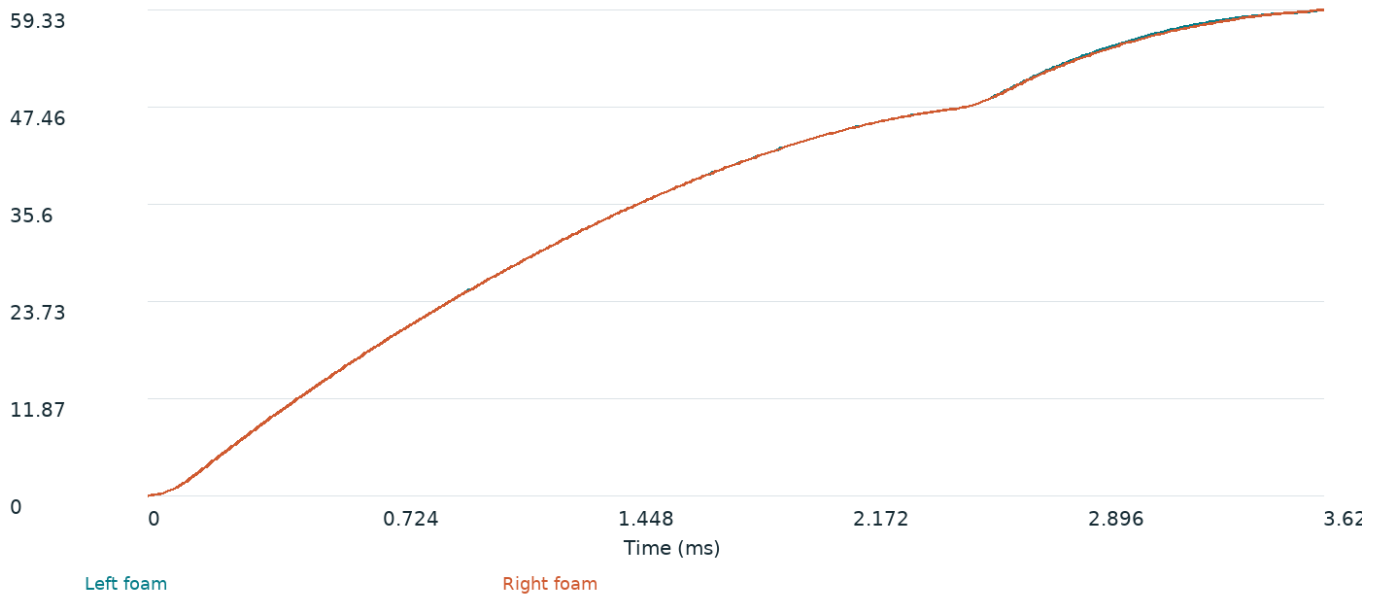
Moving assembly travel



Moving assembly travel. projects/compliant_stop_340/cases/depth6-guided-v2/assessment/time_histories.csv

Foam compression

Compression (%)



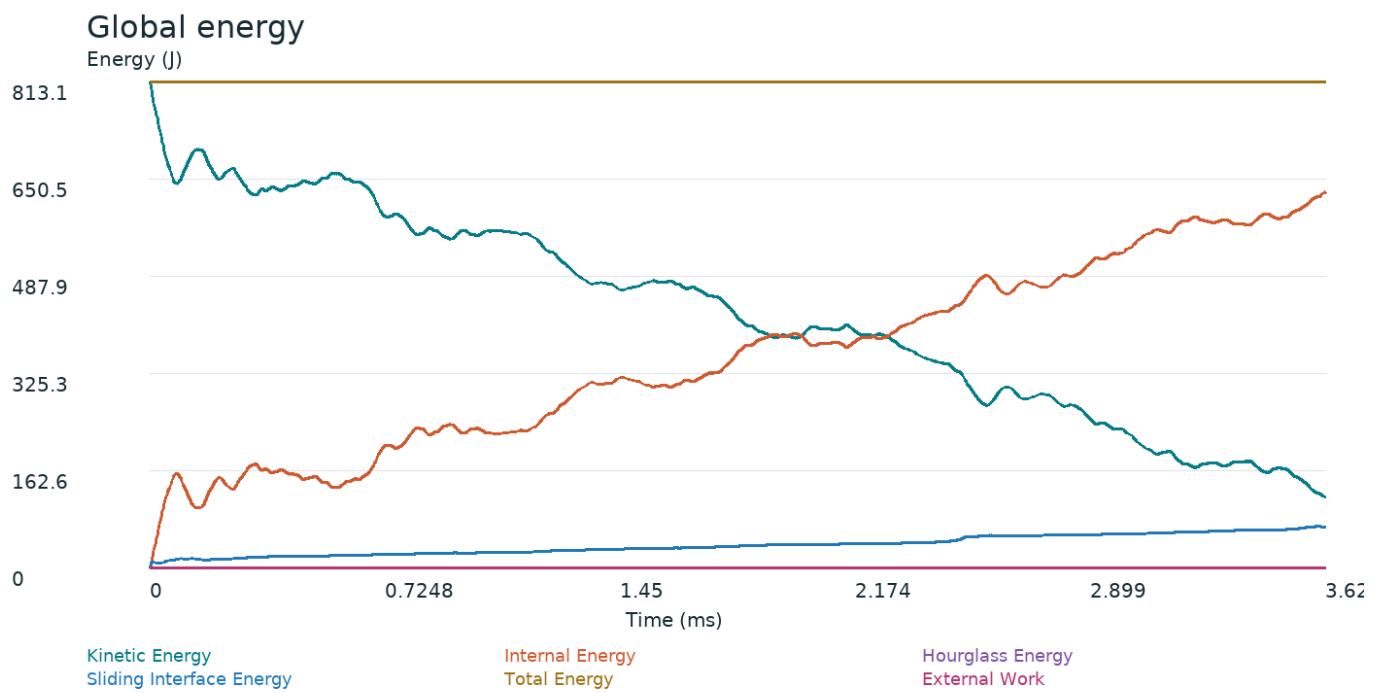
Foam compression. projects/compliant_stop_340/cases/depth6-guided-v2/assessment/time_histories.csv

Regional P99 von Mises stress

Stress (MPa)



Regional P99 von Mises stress. projects/compliant_stop_340/cases/depth6-guided-v2/assessment/stress_histories.csv



Global energy. projects/compliant_stop_340/cases/depth6-guided-v2/assessment/global_energy_histories.csv

Results

Run 013 - impact evidence

Evidence status: failed. This report describes only extracted evidence from the selected run. Numerical checks alone do not establish physical qualification.

Geometry and model setup

Recorded model item	Value
kind	impact
pusher kg	0.488741
body kg	0.41625
cable cap kg	0.0724904
motors kg	0.1616
moving kg	0.650341
initial ke J	812.926
spreader each kg	0.135871

Solved mesh

Recorded mesh source: mesh_4mm_repaired. Mesh and timestep convergence are not established by these histories.

Motion and stopping event

Signed forward velocity, travel and foam compression are measured histories. Negative forward velocity denotes rebound; a partial history cannot establish the complete stopping event.

Measured result	Value	Unit
Peak stopping force	174104	N
Maximum travel	92.0801	mm
Peak regional P99 (cap)	126.722	MPa
Last extracted time	3.61998	ms
Final forward velocity	11.2036	m/s

Regional stress and energy

Regional P99 values are elastic demand measures across saved states, not printed-part strength predictions. Force peaks and percentile peaks can occur at different times.

Numerical checks and qualification

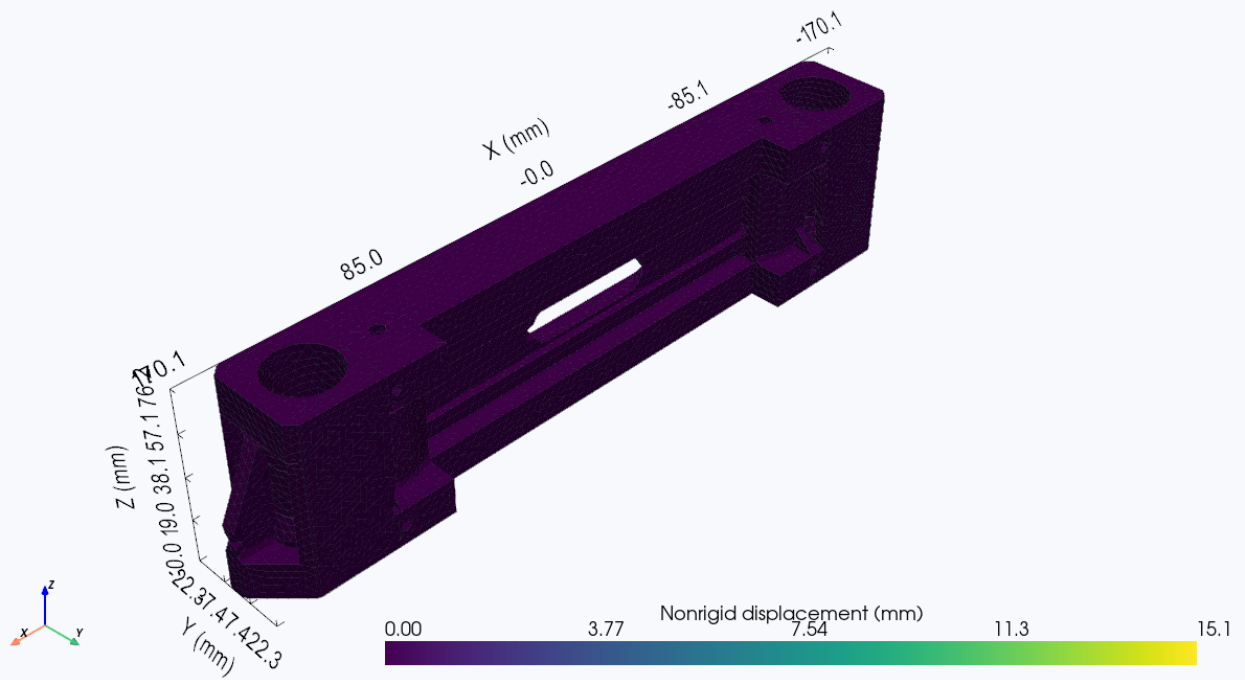
Extracted numerical check	Evidence
normal termination	false
requested time reached	false
binary matches nodout	true
moving mass consistent	true
initial energy error below 0p5pct	true
energy balance error below 5pct	true
hourglass below 5pct initial energy	true
sum abs nonfriction contact energy below 5pct initial energy	true
contact impulse momentum error below 5pct	true
no axial spc on moving or guide nodes	true
rigid guides leave only axial translation free	true
no prescribed motion	true
no detected deletion	true
positive sampled foam jacobians	true
first stop captured	false
foam compression below 80pct	true
mass scaling disabled or quantified below 0p1pct moving mass	true
all transverse guide equations satisfied and no axial equation	true

Partial extraction: these histories do not establish a completed solve.

Images, animations and source evidence

PARTIAL PREVIEW
Rigid motion removed; deformation 1x; original outline gray

depth6-guided-v2 t = 0.010 ms

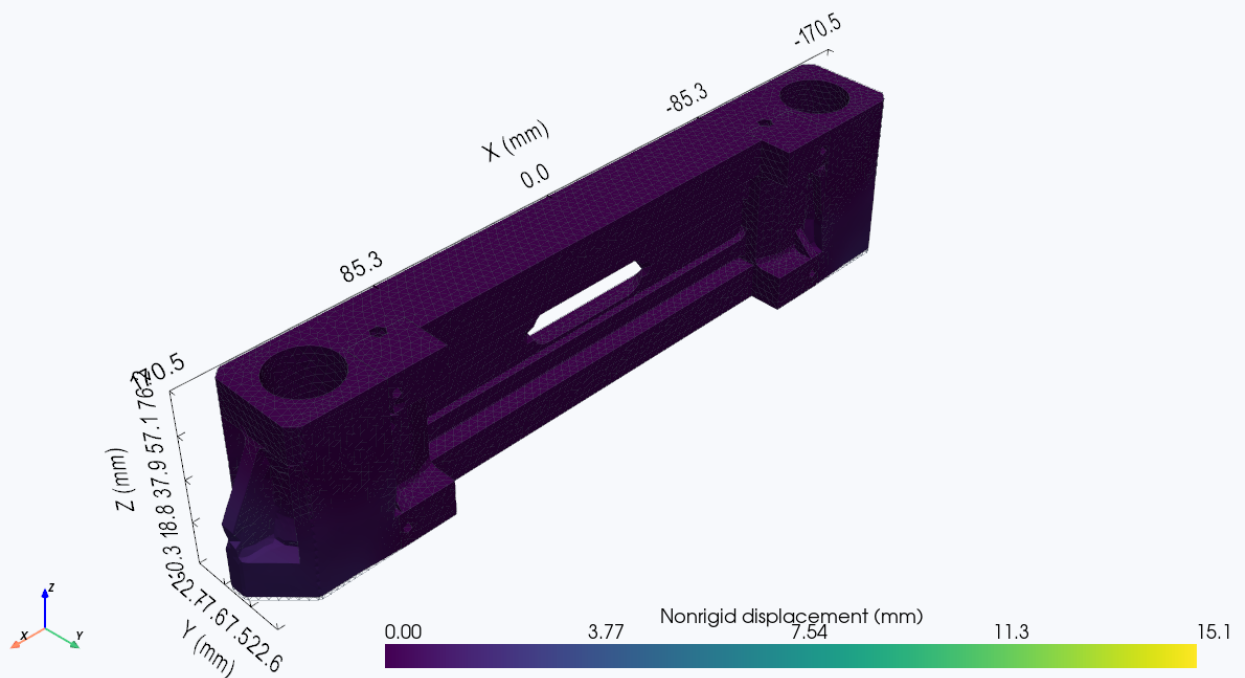


Elastic stress demand / generic foam — no failure prediction

deformation_0001_0.010ms.png

PARTIAL PREVIEW
Rigid motion removed; deformation 1x; original outline gray

depth6-guided-v2 t = 0.050 ms

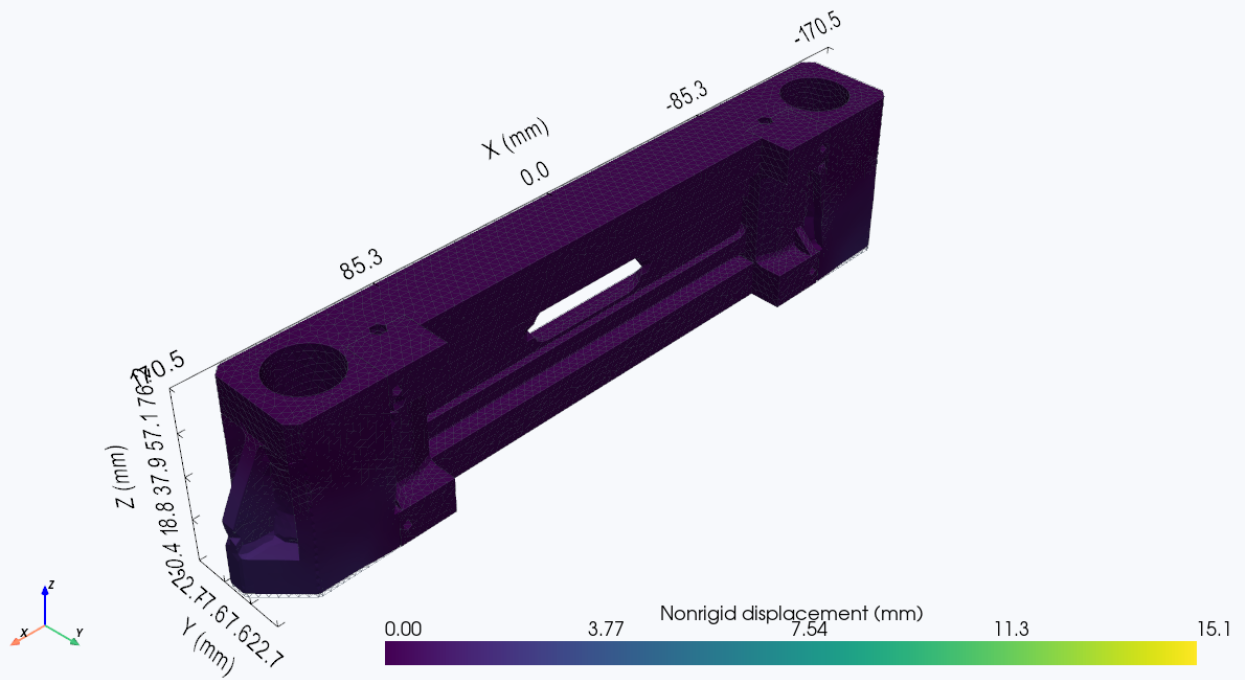


Elastic stress demand / generic foam — no failure prediction

deformation_0005_0.050ms.png

PARTIAL PREVIEW
Rigid motion removed; deformation 1x; original outline gray

depth6-guided-v2 t = 0.060 ms

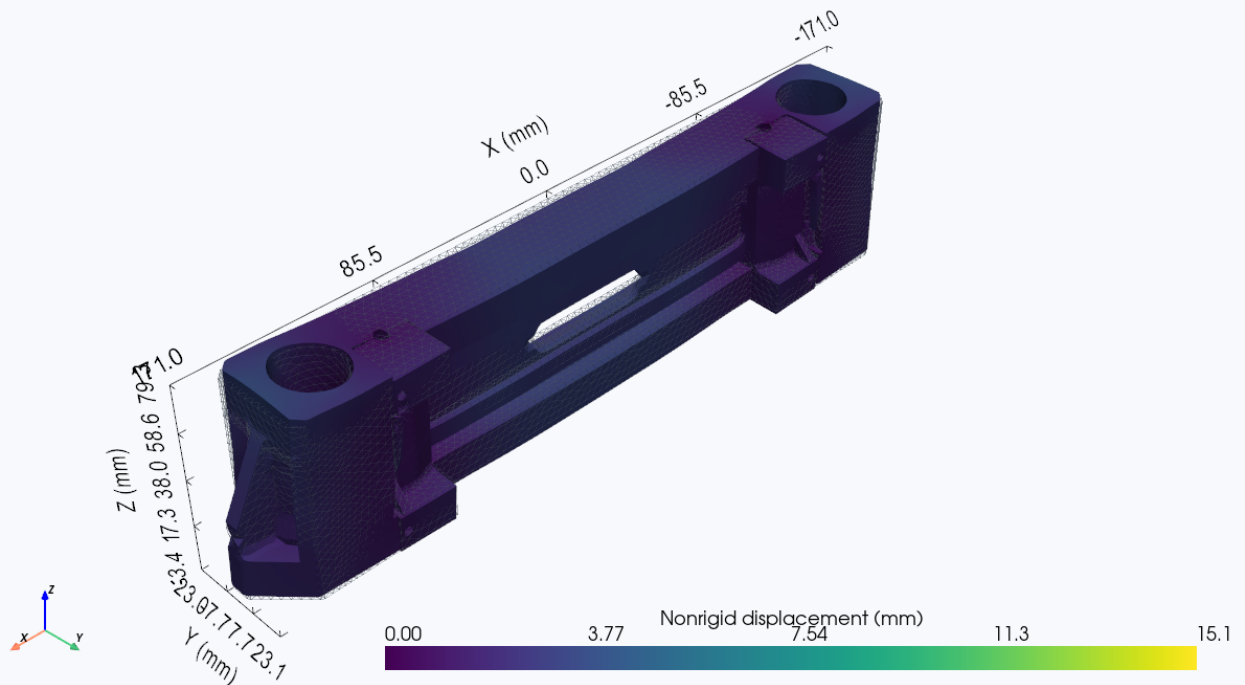


Elastic stress demand / generic foam — no failure prediction

deformation_0006_0.060ms.png

PARTIAL PREVIEW
Rigid motion removed; deformation 1x; original outline gray

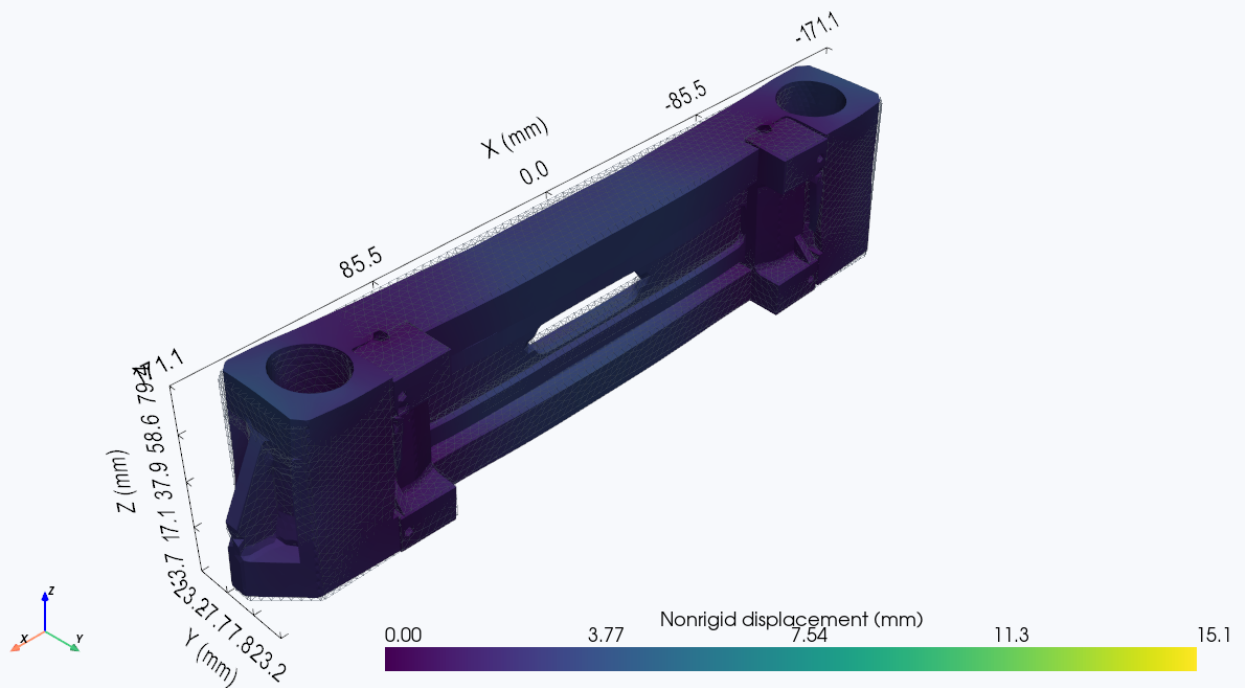
depth6-guided-v2 t = 0.210 ms



Elastic stress demand / generic foam — no failure prediction

deformation_0021_0.210ms.png

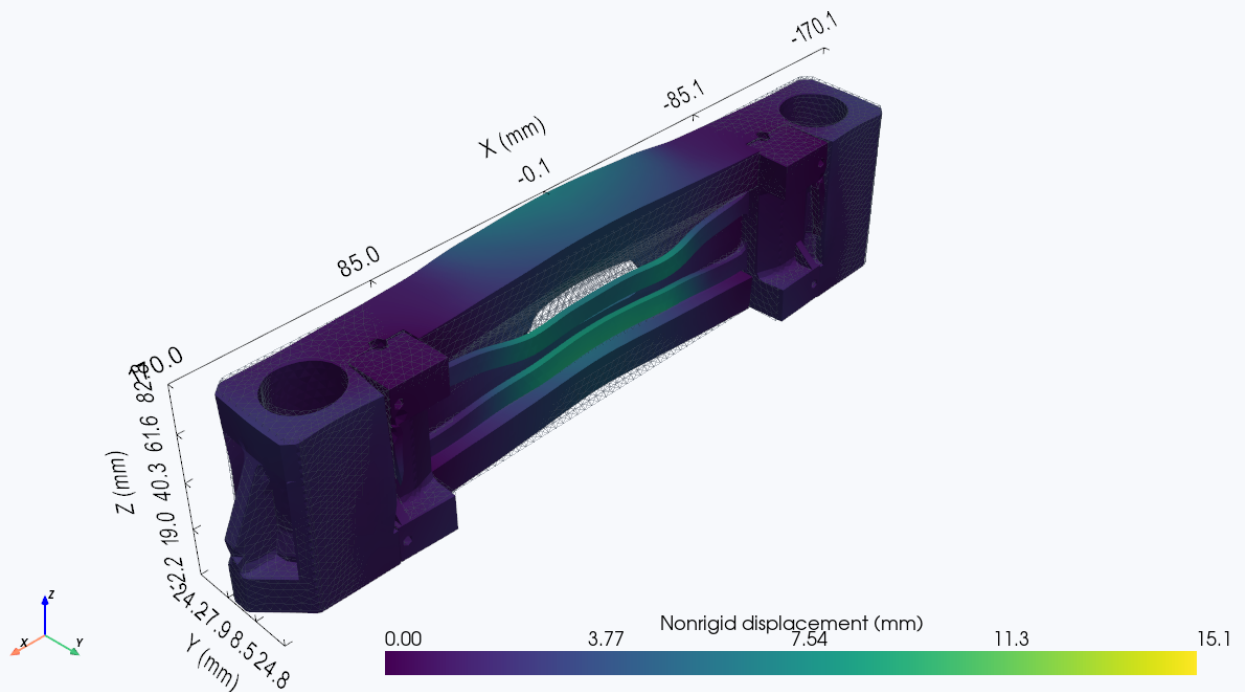
PARTIAL PREVIEW
depth6-guided-v2 t = 0.220 ms
Rigid motion removed; deformation 1x; original outline gray



Elastic stress demand / generic foam — no failure prediction

deformation_0022_0.220ms.png

PARTIAL PREVIEW
depth6-guided-v2 t = 0.820 ms
Rigid motion removed; deformation 1x; original outline gray

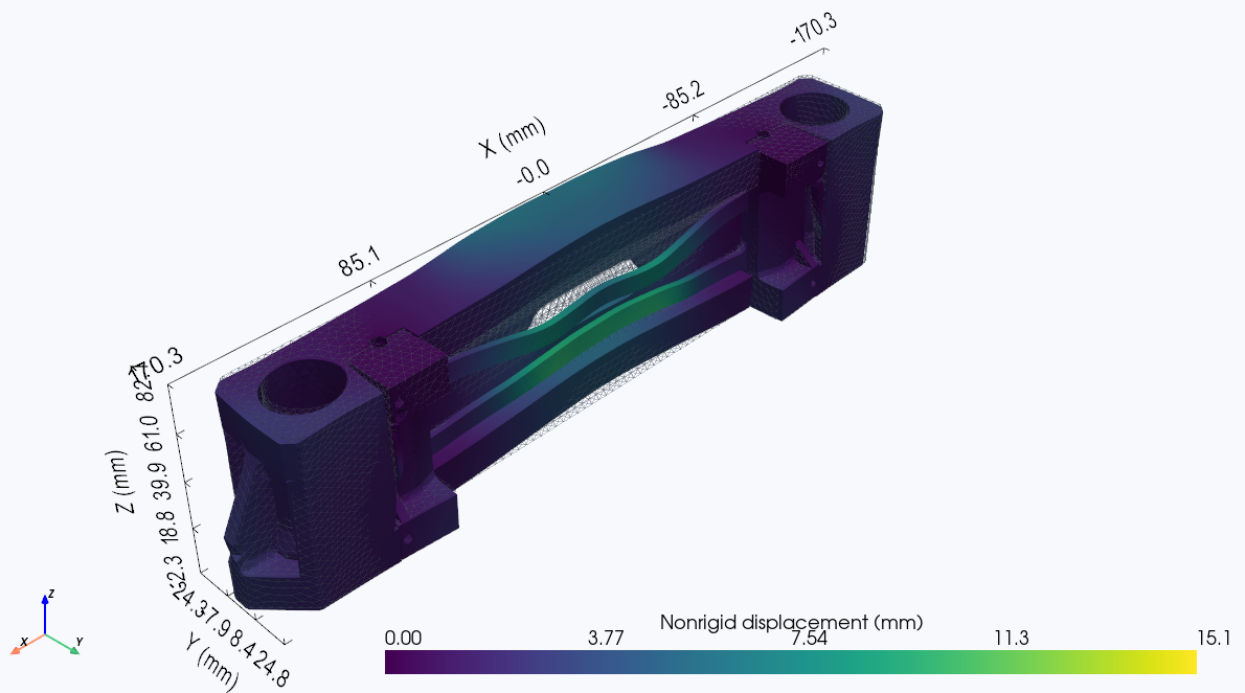


Elastic stress demand / generic foam — no failure prediction

deformation_0082_0.820ms.png

PARTIAL PREVIEW
Rigid motion removed; deformation 1x; original outline gray

depth6-guided-v2 t = 0.850 ms

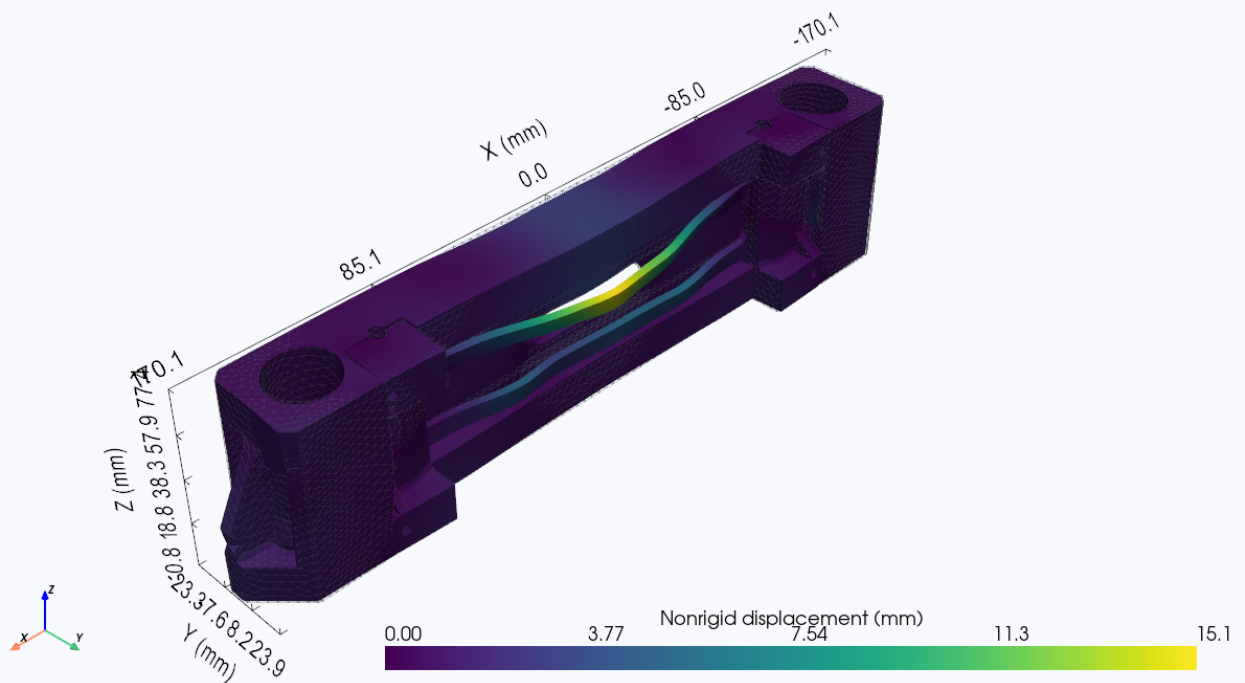


Elastic stress demand / generic foam — no failure prediction

deformation_0085_0.850ms.png

PARTIAL PREVIEW
Rigid motion removed; deformation 1x; original outline gray

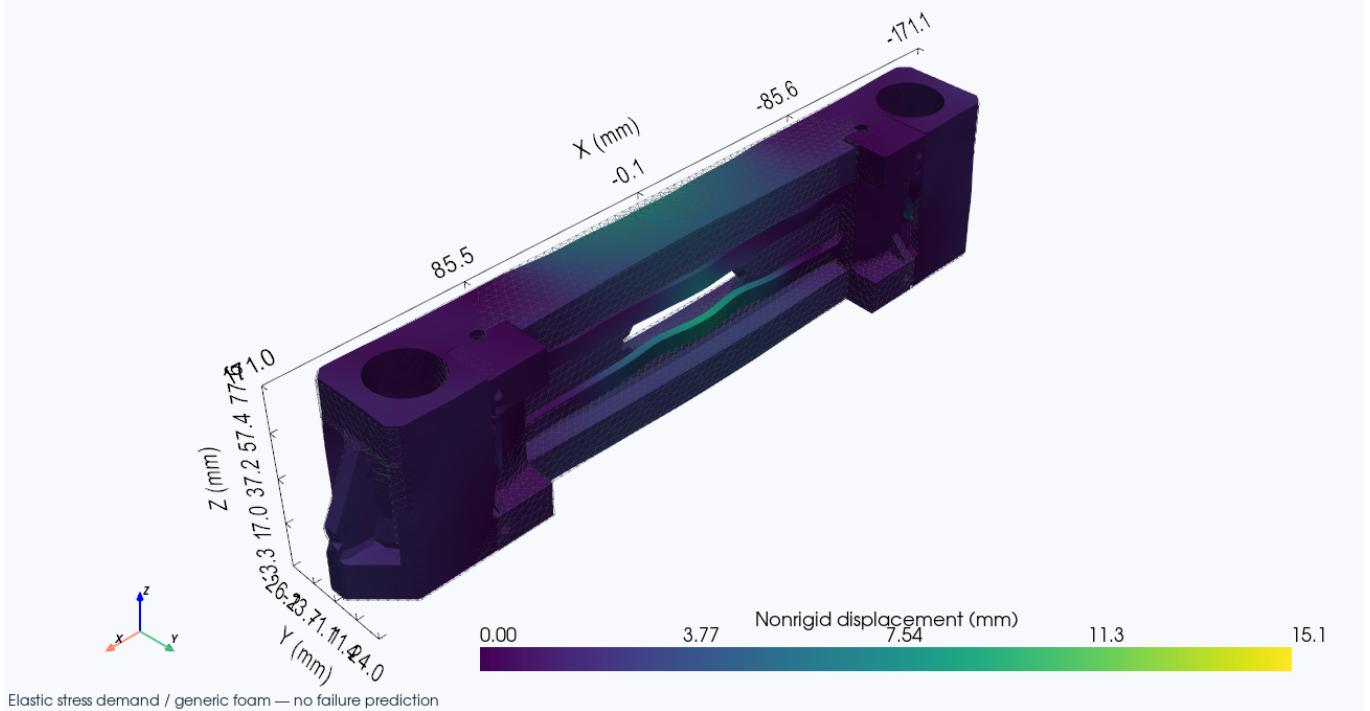
depth6-guided-v2 t = 1.710 ms



Elastic stress demand / generic foam — no failure prediction

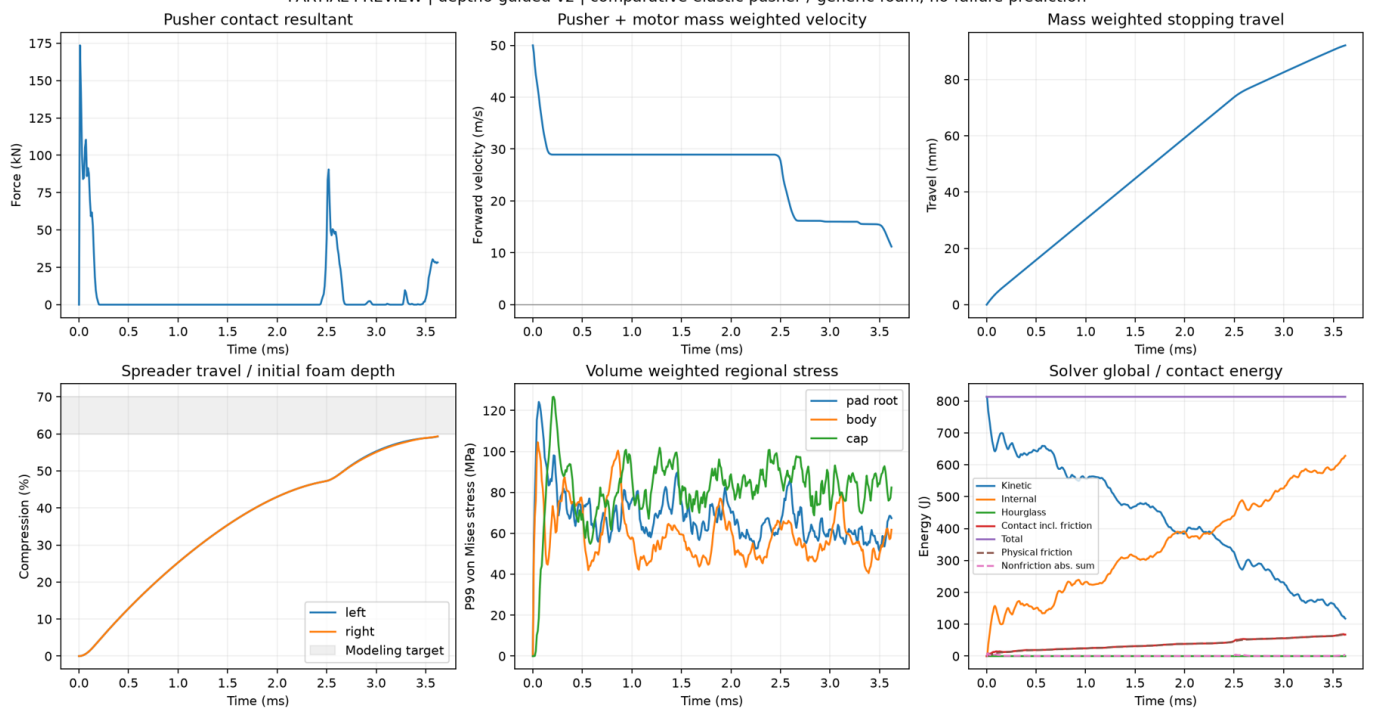
deformation_0171_1.710ms.png

Rigid motion removed; deformation 1x; original outline gray



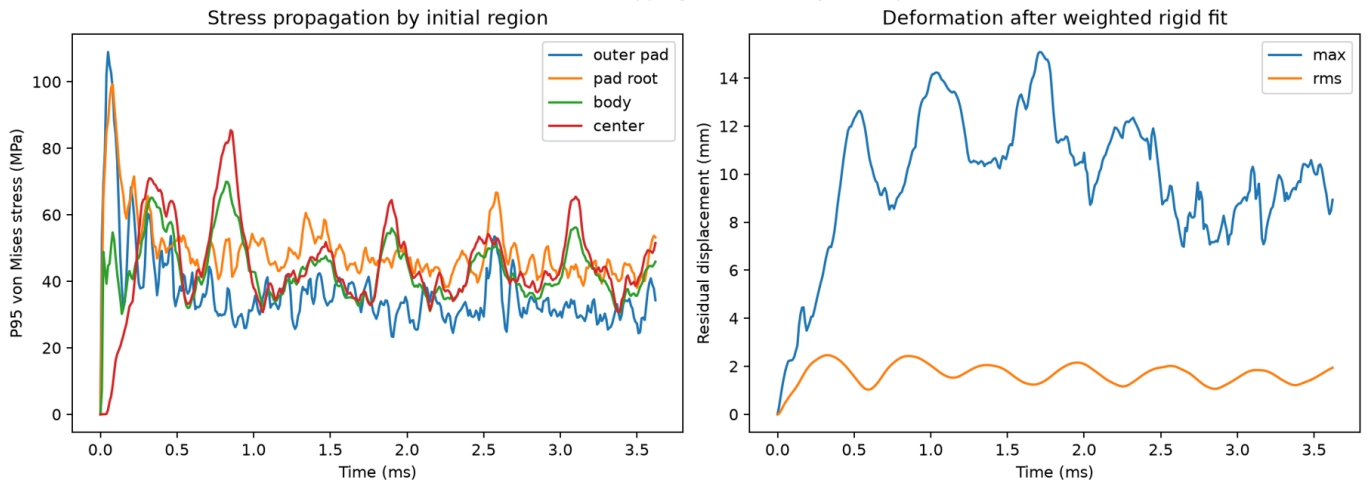
deformation_0362_3.620ms.png

PARTIAL PREVIEW | depth6-guided-v2 | comparative elastic pusher / generic foam; no failure prediction



histories.png

PARTIAL PREVIEW: stopping event is not yet complete

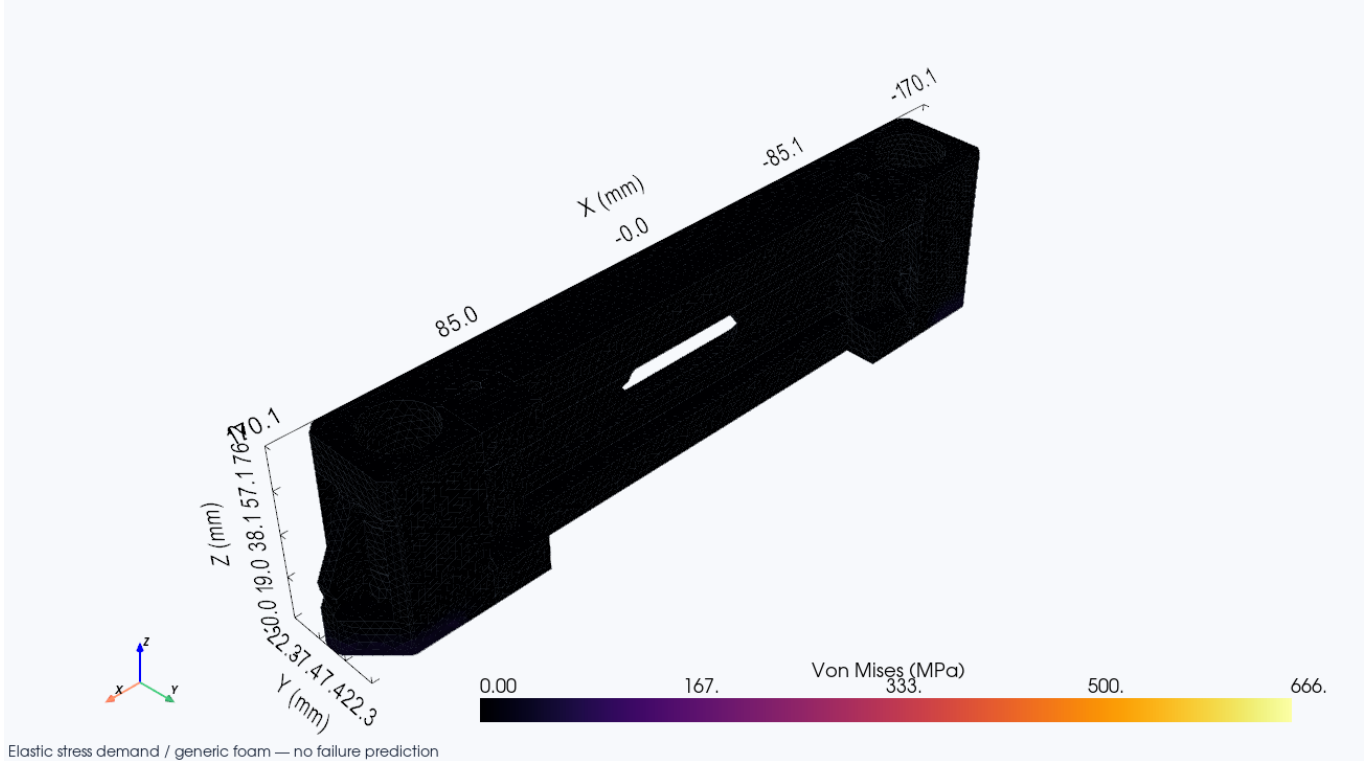


stress-propagation-deformation.png

PARTIAL PREVIEW

depth6-guided-v2 t = 0.010 ms

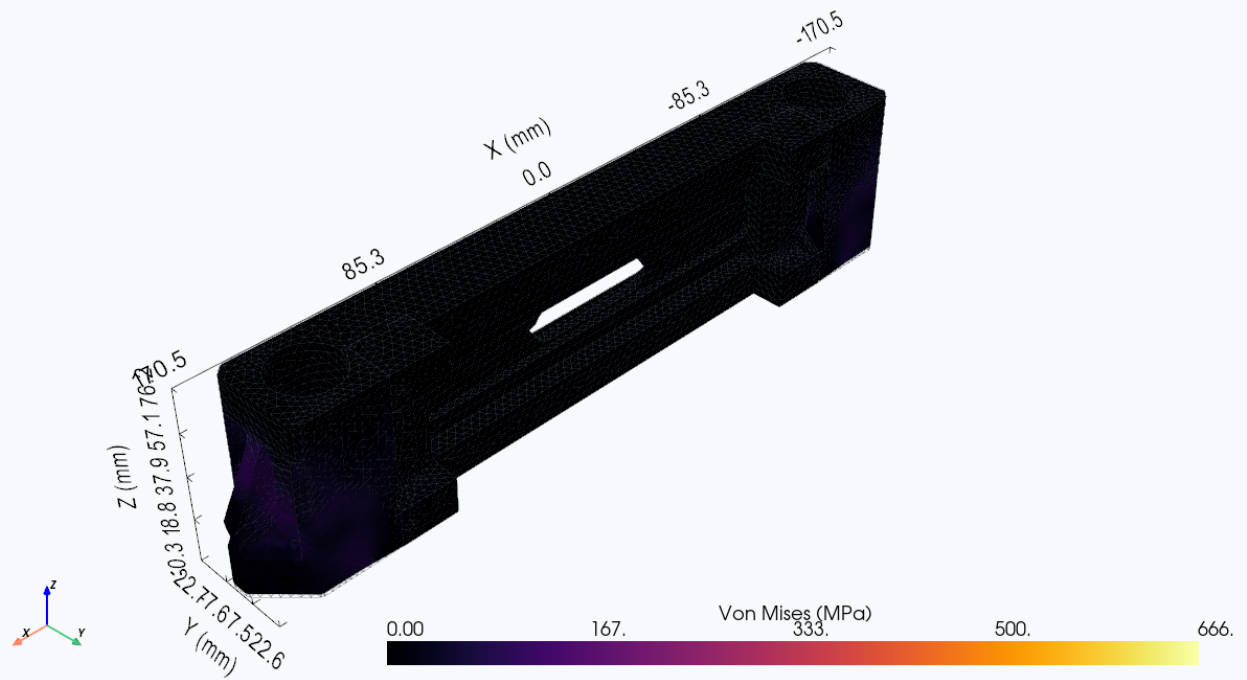
Rigid motion removed; deformation 1x; original outline gray



Elastic stress demand / generic foam — no failure prediction

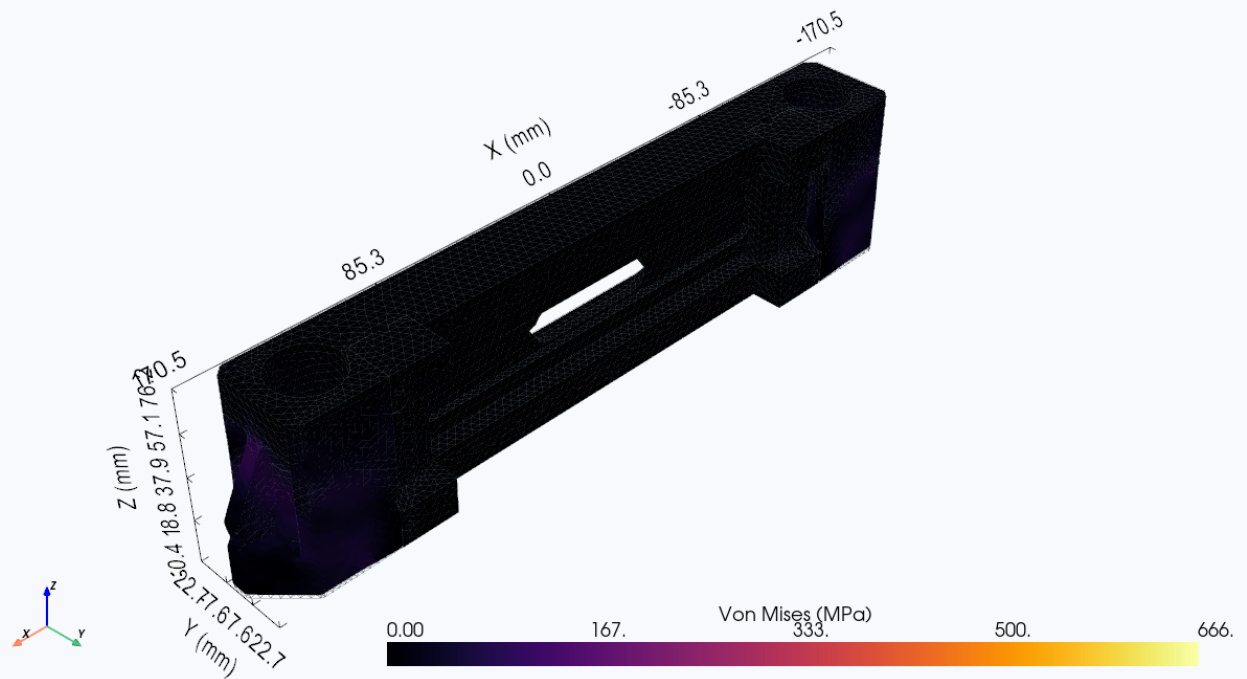
stress_0001_0.010ms.png

PARTIAL PREVIEW depth6-guided-v2 t = 0.050 ms
Rigid motion removed; deformation 1x; original outline gray



stress_0005_0.050ms.png

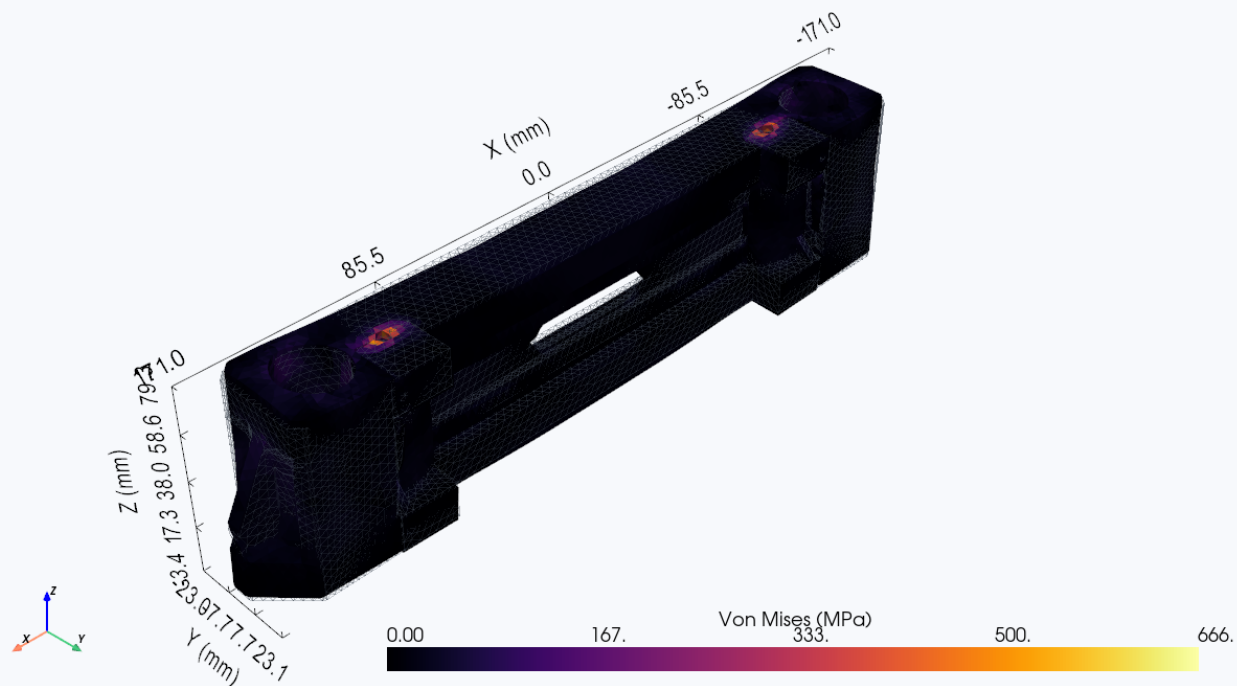
PARTIAL PREVIEW depth6-guided-v2 t = 0.060 ms
Rigid motion removed; deformation 1x; original outline gray



stress_0006_0.060ms.png

PARTIAL PREVIEW
Rigid motion removed; deformation 1x; original outline gray

depth6-guided-v2 t = 0.210 ms

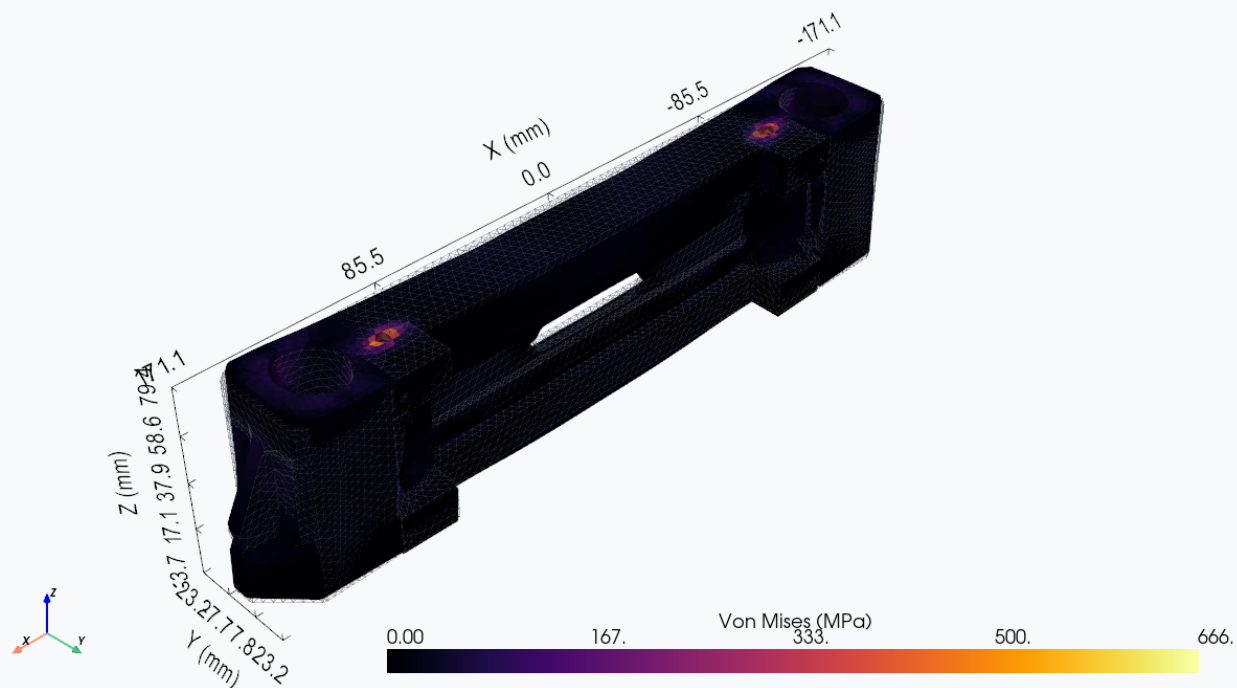


Elastic stress demand / generic foam — no failure prediction

stress_0021_0.210ms.png

PARTIAL PREVIEW
Rigid motion removed; deformation 1x; original outline gray

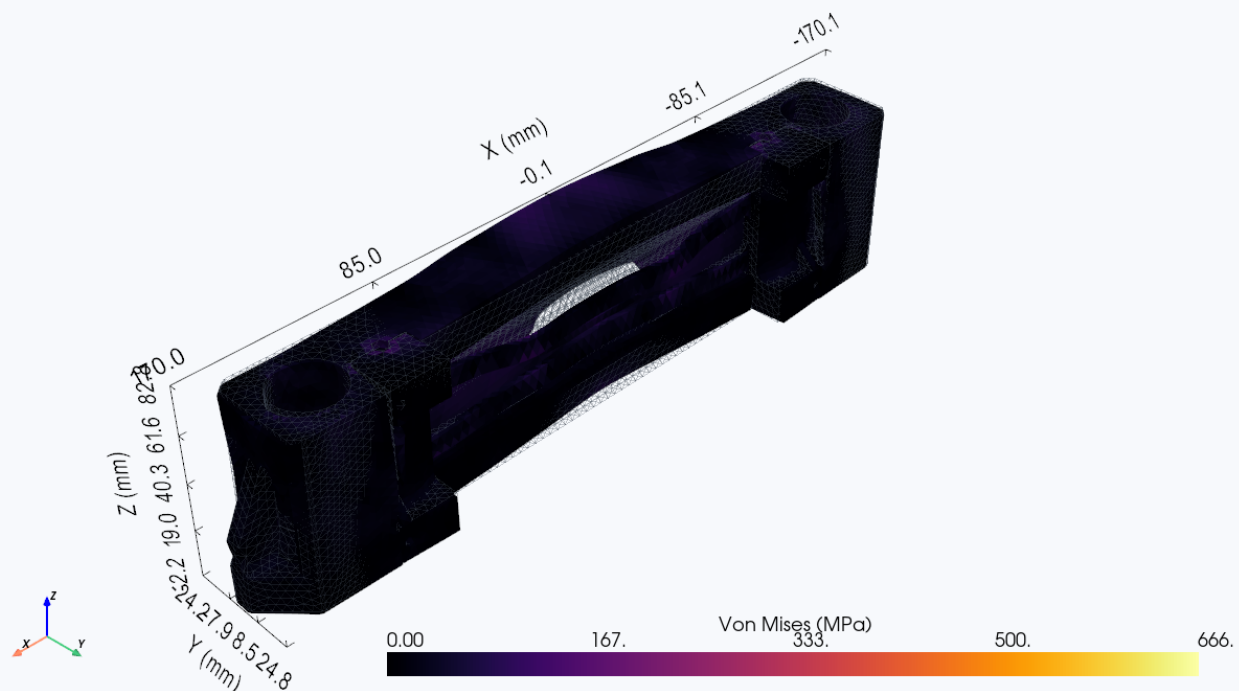
depth6-guided-v2 t = 0.220 ms



Elastic stress demand / generic foam — no failure prediction

stress_0022_0.220ms.png

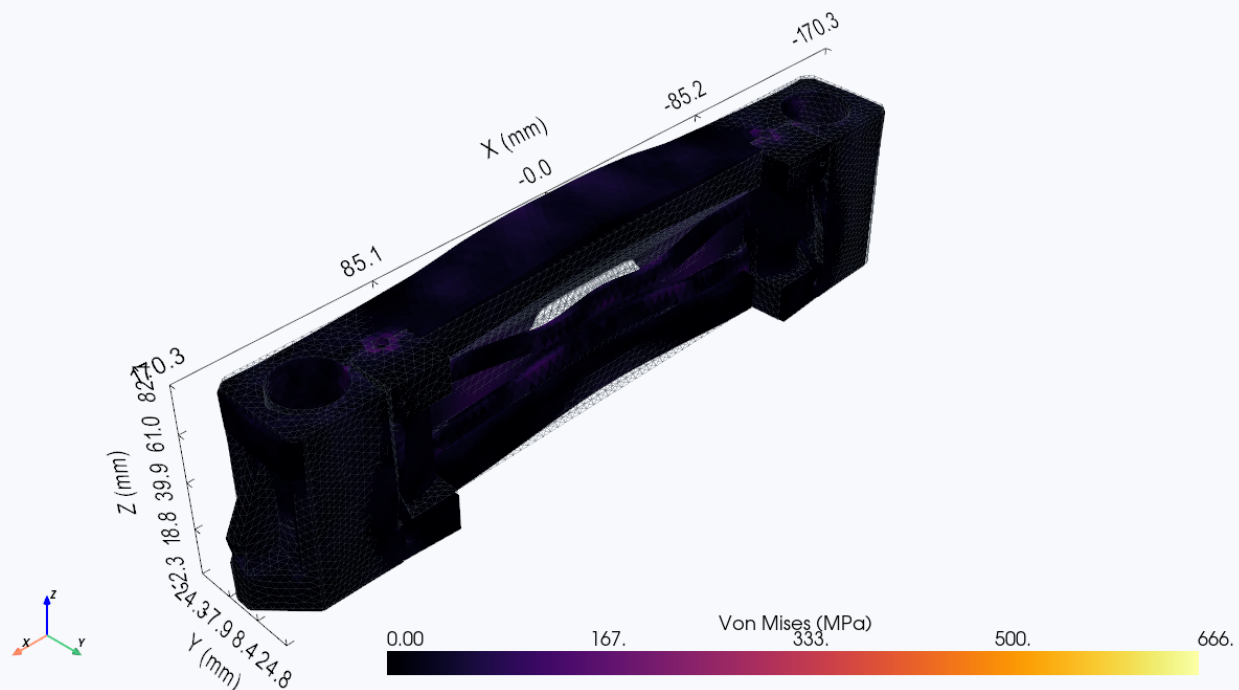
PARTIAL PREVIEW depth6-guided-v2 t = 0.820 ms
Rigid motion removed; deformation 1x; original outline gray



Elastic stress demand / generic foam — no failure prediction

stress_0082_0.820ms.png

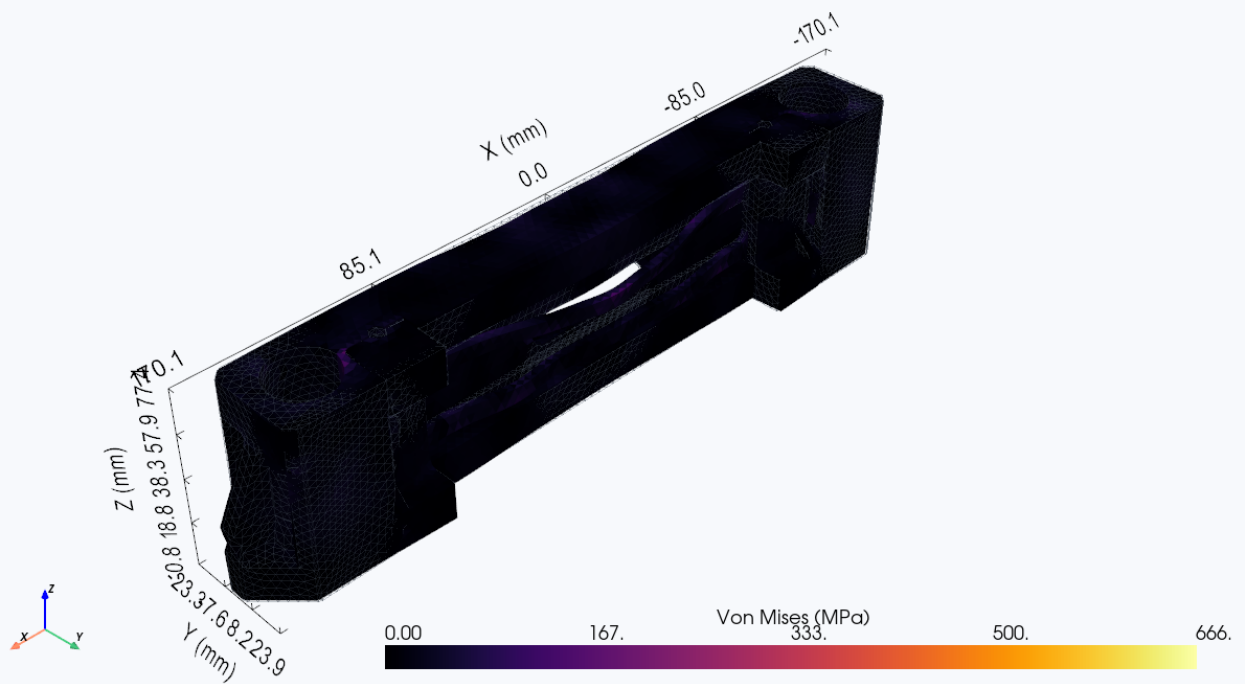
PARTIAL PREVIEW depth6-guided-v2 t = 0.850 ms
Rigid motion removed; deformation 1x; original outline gray



Elastic stress demand / generic foam — no failure prediction

stress_0085_0.850ms.png

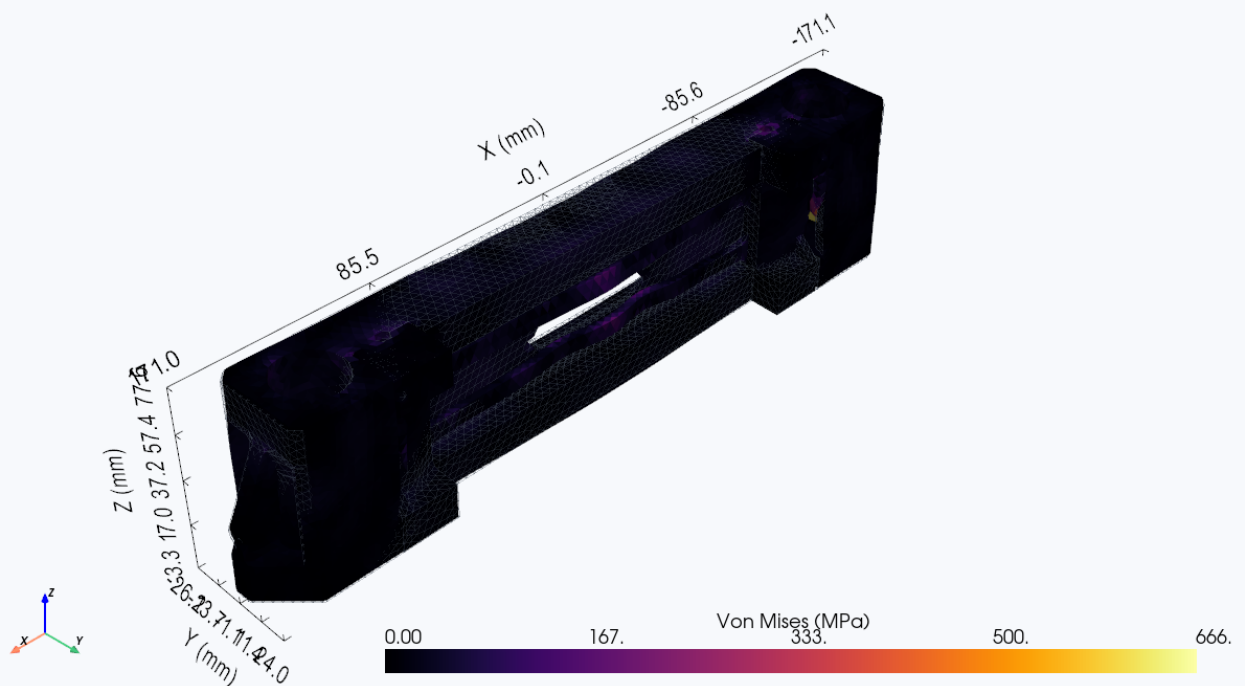
PARTIAL PREVIEW depth6-guided-v2 t = 1.710 ms
Rigid motion removed; deformation 1x; original outline gray



Elastic stress demand / generic foam — no failure prediction

stress_0171_1.710ms.png

PARTIAL PREVIEW depth6-guided-v2 t = 3.620 ms
Rigid motion removed; deformation 1x; original outline gray



Elastic stress demand / generic foam — no failure prediction

stress_0362_3.620ms.png

- cable_guide_motion.csv
- contact_energy_by_interface.csv

- contact_energy_components.csv
- contact_histories.csv
- failure_geometry_diagnostic.json
- global_energy_histories.csv
- metrics.json
- part_energy_histories.csv
- sampled_planar_penetration.csv
- spc_histories.csv
- stress_histories.csv
- time_histories.csv
- weighted_guide_constraints.csv

Generated from selected-run extracted evidence; no authored report is available. Animation figures and their captions are excluded from this PDF; animations remain available in the standalone HTML.